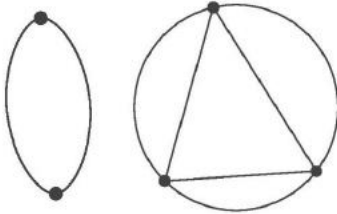


Given: Due:	Further Maths (Decision) HW 05	MARK SCHEME
Chapter 2: Graph theory		
(a) Explain why it is not possible to draw a graph with exactly 5 nodes with orders 1, 3, 4, 4 and 5		
(1)		
A connected graph has exactly 5 nodes and contains 18 arcs. The orders of the 5 nodes are $2^{2x} - 1$, 2^x , $x + 1$, $2^{x+1} - 3$ and $11 - x$		
(b) (i) Calculate x .		
(ii) State whether the graph is Eulerian, semi-Eulerian or neither. You must justify your answer.		
(6)		
(c) Draw a graph which satisfies all of the following conditions:		
• The graph has exactly 5 nodes. • The nodes have orders 2, 2, 4, 4 and 4 • The graph is not Eulerian.		
(2)		

Question	Scheme	Marks	AOs
(a)	E.g. a graph cannot contain an odd number of odd nodes E.g. number of arcs = $\frac{1+3+4+4+5}{2} = 8.5 \notin \mathbb{Z}$	B1	2.4
		(1)	
(b)(i)	$(2^{2x} - 1) + (2^x) + (x + 1) + (2^{x+1} - 3) + (11 - x) = 2(18)$	M1	1.1b
	$2^{2x} + 3(2^x) - 28 = 0 \Rightarrow x = \dots$	M1	1.1b
	$(2^x + 7)(2^x - 4) = 0 \Rightarrow x = 2$	A1	1.1b
		(3)	
(b)(ii)	The order of the nodes are 9, 15, 3, 4, 5	M1	2.1
	Therefore the graph is neither Eulerian nor semi-Eulerian as there are more than two odd nodes	A1	2.4
		A1	2.2a
		(3)	
(c)		M1	2.5
		A1	2.2a
		(2)	

(9 marks)

Notes:

(a)	B1: explanation referring to need for an even number of odd nodes oe
(b)	M1: forming an equation involving the orders of the 5 odd nodes and $2(18)$ M1: simplifies to a quadratic in 2^x and attempts to solve A1: 2 cao M1: construct an argument involving the order of the 5 nodes A1: explanation considering the number of odd nodes A1: deduction that therefore it is neither Eulerian nor semi-Eulerian
(c)	M1: interprets mathematical language to construct a disconnected graph A1: deduce a correct graph



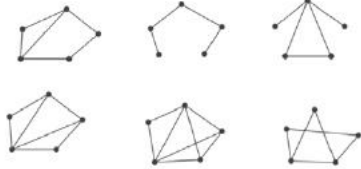
- (a) Draw the graph K_5 (1)
- (b) (i) In the context of graph theory explain what is meant by 'semi-Eulerian'.
(ii) Draw two semi-Eulerian subgraphs of K_5 , each having five vertices but with a different number of edges. (3)
- (c) Explain why a graph with exactly five vertices with vertex orders 1, 2, 2, 3 and 4 cannot be a tree. (2)

As a 'new specification' question, this was met with varying degrees of success on the part of candidates. There were some perfect or near perfect responses but also many responses which lacked understanding and/or knowledge. While most were able to draw K_5 in part (a), others were clearly confused by the terminology. Some drew a pentagon, others drew K_6 . Some candidates missed out a single arc and many other subgraphs of K_5 appeared here.

In part (b)(i), most candidates were familiar with the term 'semi-Eulerian', however often definitions fell short of the rigour required. It was common to see comments which did not imply 'exactly' two nodes of odd order. Examiners often commented that the phrase 'at least two odd nodes' was seen on a number of occasions. Many candidates also provided non credit-worthy descriptions involving the ability to 'traverse each arc'.

Part (b)(ii) was often well answered, even when part (a) had not been. Most candidates drew semi-Eulerian subgraphs of K_5 but were sometimes let down by both graphs having the same number of arcs. Others incorrectly drew graphs with multiple arcs between pairs of nodes and some drew graphs with fewer than 5 nodes. Examiners noted that a few Eulerian graphs were provided here as well as some non-semi-Eulerian graphs with multiple nodes of order 1.

Part (c) discriminated well. It was common to see no response here and extremely common to see incorrect attempts. Nonetheless, the most able candidates were able to provide coherent and concise arguments which focussed on the key issues – most usually, the number of arcs required for a tree with the stated orders compared to the number of arcs on a tree with five nodes. Examiners noted that some candidates gave arguments based on the direct connection between the vertex of order 4 and each of the other vertices together with the resulting need for all other nodes to have order 1 for a tree. Other valid arguments were rarer but also appeared from time to time. For some candidates, there was a lot of confusion between arcs and nodes and many stated that there must be 6 nodes on a graph with the stated orders. Other candidates incorrectly deduced that 'because the graph is Semi-Eulerian' it could not be a tree, and some candidates seemed to think that there was only one odd node here. There were a substantial number of candidates who were able to pick up one mark out of two here for making some progress towards a correct explanation.

Question	Scheme	Marks	AOs
(a)		B1	1.2
		(1)	
(b)(i)	A semi-Eulerian graph contains <u>exactly two nodes of odd order</u> (and any number of nodes of even order)	B1	2.5
(b)(ii)	e.g. (two semi-Eulerian subgraphs of K_5 with a different number of edges) 	B1 B1	1.1b 1.1b
		(3)	
(c)	e.g. The graph with five vertices has $\frac{1+2+2+3+4}{2} = 6$ arcs but a tree on five nodes would contain only 4 arcs	B1 B1dep	2.2a 2.4
		(2)	
(6 marks)			

Notes
(a) B1: CAO (give bod for position of nodes) (b)(i) B1: CAO (accept 'there are exactly two odd nodes' but must contain exact oe (e.g. 'only two odd nodes' or 'all but 2 nodes have an even order' but not 'the graph has two odd nodes')) (b)(ii) B1: One correct semi-Eulerian subgraph of K_5 with five nodes B1: Two correct semi-Eulerian subgraphs of K_5 with five nodes – note that the graphs must have a different number of edges (c) B1: Deducing that the graph has 6 arcs or a tree on five nodes has 4 arcs or the node of order 4 must be connected to the other 4 nodes or an argument based on the sum of the orders of both the graph and the tree (but must relate the orders to the number of arcs and not the number of nodes) or the node with order 4 and one of the nodes of orders 2 or 3 would create a cycle or a tree must have two nodes of order 1 B1dep: Complete argument – graph has 6 arcs and the tree would only have 4 arcs or the sum of the orders is 12 compared to 8 for the tree or the node of order 4 must be connected to the other 4 nodes therefore all the other vertices would have to have order 1 or the graph has 6 arcs and therefore with 5 vertices there would have to be cycles or the node of order 4 is connected to the other 4 nodes and so together with the node of order 3 (or 2) a cycle would be formed or a tree must have at least two nodes of order 1 as otherwise a cycle would be formed Note: no marks in (c) for attempts based only on examples of graphs drawn with the vertex orders as stated